

Scoring Rules, Generalized Entropy, and Utility Maximization

Victor Richmond Jose, Robert Nau, & Robert Winkler
Fuqua School of Business
Duke University

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Overview

- **Scoring rules** are reward functions for defining subjective probabilities and eliciting them in forecasting applications and experimental economics (de Finetti, Brier, Savage, Selten...)
- **Cross-entropy**, or **divergence**, is a physical measure of information gain in communication theory and machine learning (Shannon, Kullback-Leibler...)
- **Utility maximization** is the decision maker's objective in Bayesian decision theory and game theory (von Neumann & Morgenstern, Savage...)

General connections

- Any decision problem under uncertainty may be used to define a scoring rule or measure of divergence between probability distributions.
- The expected score or divergence is merely the expected-utility gain that results from solving the problem using the decision maker's "true" (or posterior) probability distribution $\mathbf{p}$ rather than some other "baseline" (or prior) distribution $\mathbf{q}$.
- These connections have been of interest in the recent literature of robust Bayesian inference and mathematical finance.

Specific results

- We explore the connections among the best-known parametric families of generalized scoring rules, divergence measures, and utility functions.
- The expected scores obtained by truthful probability assessors turn out to correspond exactly to well-known generalized divergences.
- They also correspond exactly to expected-utility gains in financial investment problems with utility functions from the linear-risk-tolerance (a.k.a. HARA) family.
- These results generalize to incomplete markets via a primal-dual pair of convex programs.

Part 1: Scoring rules

- Consider a probability forecast for a discrete event with n possible outcomes (“states of the world”).
 - Let $\mathbf{e}_i = (0, \dots, 1, \dots, 0)$ denote the indicator vector for the i^{th} state (where 1 appears in the i^{th} position).
 - Let $\mathbf{p} = (p_1, \dots, p_n)$ denote the forecaster’s *true* subjective probability distribution over states.
 - Let $\mathbf{r} = (r_1, \dots, r_n)$ denote the forecaster’s *reported* distribution (if different from $\mathbf{p}$).
- (Later, let $\mathbf{q} = (q_1, \dots, q_n)$ denote a *baseline* distribution upon which the forecaster seeks to improve.)

Definition of a scoring rule

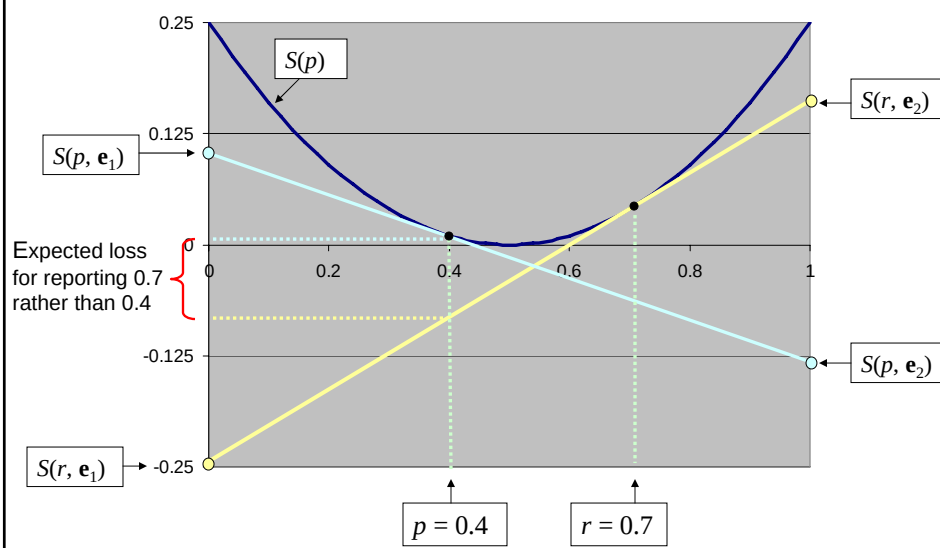
- A scoring rule is a function $S(\mathbf{r}, \mathbf{p})$, linear in $\mathbf{p}$, that determines the forecaster’s *expected* score (reward) for reporting $\mathbf{r}$ when her true distribution is $\mathbf{p}$.
- The actual score is $S(\mathbf{r}, \mathbf{e}_i)$ when the i^{th} state occurs.
- $S(\mathbf{p}) \equiv S(\mathbf{p}, \mathbf{p})$ will denote the forecaster’s expected score for truthfully reporting her true distribution $\mathbf{p}$

Proper scoring rules

- The scoring rule S is [strictly] proper if $S(\mathbf{p}) \geq [>] S(\mathbf{r}, \mathbf{p})$ for all $\mathbf{r} [\neq \mathbf{p}]$, i.e., if the forecaster's expected score is [uniquely] maximized when she reports her true probabilities.
- S is [strictly] proper iff $S(\mathbf{p})$ is a [strictly] convex function of $\mathbf{p}$.
- If S is strictly proper, then it is uniquely determined from $S(\mathbf{p})$ by McCarthy's (1956) formula:

$$S(\mathbf{r}, \mathbf{p}) = S(\mathbf{r}) + \nabla S(\mathbf{r}) \cdot (\mathbf{p} - \mathbf{r})$$

Construction of scoring rule from $S(p)$



Common scoring rules

The three most commonly used scoring rules are:

- The *quadratic* scoring rule:

$$S(\mathbf{p}, \mathbf{e}_i) = - (\|\mathbf{e}_i - \mathbf{p}\|_2)^2$$

Score is squared Euclidean distance between $\mathbf{p}$ and $\mathbf{e}_i$

- The *spherical* scoring rule:

$$S(\mathbf{p}, \mathbf{e}_i) = p_i / \|\mathbf{p}\|_2$$

Score vector lies on the surface of a pseudosphere centered at the origin

- The *logarithmic* scoring rule:

$$S(\mathbf{p}, \mathbf{e}_i) = \ln(p_i)$$

History of common scoring rules

- The quadratic scoring rule was introduced by de Finetti (1937, 1974) to define subjective probability; later used by Brier (1950) as a tool for evaluating and paying weather forecasters; more recently used to reward subjects in economic experiments.
- Selten (1998) has presented an axiomatic argument in favor of the quadratic rule.
- The spherical and logarithmic rules were introduced by I.J. Good (1971), who also noted that the spherical and quadratic rules could be generalized to an arbitrary positive power β other than 2, leading to...

Generalized scoring rules

- *Power* scoring rule ($\rightarrow$ quadratic at $\beta = 2$):

$$\beta p_i^{\beta-1} - (\beta - 1) \left(\|\mathbf{p}\|_\beta \right)^\beta$$

- *Pseudospherical* scoring rule ($\rightarrow$ spherical at $\beta = 2$)

$$\left(\frac{p_i}{\|\mathbf{p}\|_\beta} \right)^{\beta-1}$$

Score vectors lie on the surface of a pseudosphere centered at the origin

- Both rules are valid only for $\beta > 1$ and converge to (multiples of) the logarithmic rule at $\beta = 1$.
- Under both rules, the payoff profile (risk profile) is an affine function of $p_i^{\beta-1}$

Uniform “baseline” distribution?

- The standard scoring rules are symmetric across states:
 - Payoffs in different states are ranked in order of p_i
 - The optimal expected score is minimized when $\mathbf{p}$ is the uniform distribution
 - Hence these rules implicitly reward the forecaster for departures from a uniform distribution
- But is the uniform distribution the appropriate baseline against which to measure the value of a forecast?

Rationale for a non-uniform baseline

- In nearly all applications outside of laboratory experiments, the relevant baseline is *not* uniform:
 - Weather forecasting
 - Economic forecasting
 - Technological forecasting
 - Demand for new products
 - Financial markets
 - Sports betting
- We therefore propose that the score be should be “weighted” by a non-uniform baseline distribution $\mathbf{q}$ s.t. the optimal expected score is minimized at $\mathbf{p} = \mathbf{q}$, i.e., when $\mathbf{p}$ is *uninformative with respect to $\mathbf{q}$* .

How should the dependence on a baseline distribution be modeled?

- We propose that the scoring rule should rank payoffs in order of p_i/q_i , i.e., the *relative*, not absolute, deviation of p_i from q_i .
- Rationales for this form of dependence:
 - A \$1 bet on state i at odds determined by q_i has an expected payoff of p_i/q_i , hence relative probabilities are what matter for purposes of betting.
 - Payoffs ought not to depend on the outcomes of independent events that have the same probabilities under $\mathbf{p}$ and $\mathbf{q}$, and this also constrains the payoffs to depend only on the ratio p_i/q_i .

Weighted scoring rules

- The power and pseudospherical rules can be weighted by an arbitrary baseline distribution $\mathbf{q}$ merely by replacing $p_i^{\beta-1}$ with $(p_i/q_i)^{\beta-1}$ in the formulas that determine the profiles of payoffs.
- They can also be normalized so as to be valid for all real β and to yield a score of zero in all states iff $\mathbf{p} = \mathbf{q}$, so that the *expected* score is *positive* iff $\mathbf{p} \neq \mathbf{q}$.
- The weighted rules thus measure the *information value* of knowing that the distribution is $\mathbf{p}$ rather than $\mathbf{q}$ as seen from the forecaster's perspective.

With this weighting and normalization, the power and pseudospherical rules become:

- The *weighted power* scoring rule:

$$S_{\beta}^{\mathbf{P}}(\mathbf{p}, \mathbf{e}_i \| \mathbf{q}) \equiv \frac{(p_i/q_i)^{\beta-1} - 1}{\beta - 1} - \frac{E_{\mathbf{p}}[(\mathbf{p}/\mathbf{q})^{\beta-1}] - 1}{\beta}$$

- The *weighted pseudospherical* scoring rule:

$$S_{\beta}^{\mathbf{S}}(\mathbf{p}, \mathbf{e}_i \| \mathbf{q}) \equiv \frac{1}{\beta - 1} \left(\left(\frac{p_i/q_i}{(E_{\mathbf{p}}[(\mathbf{p}/\mathbf{q})^{\beta-1}])^{1/\beta}} \right)^{\beta-1} - 1 \right)$$

Properties of weighted scoring rules

- Both rules are strictly proper for *all real* β .
- Both rules $\rightarrow$ *weighted logarithmic rule* $\ln(p_i/q_i)$ at $\beta=1$.
- For the same $\mathbf{p}$, $\mathbf{q}$, and β , the vector of weighted power scores is an affine transformation of the vector of weighted pseudospherical scores, since both are affine functions of $(p_i/q_i)^{\beta-1}$.
- However, the two rules present different incentives for information-gathering and honest reporting.
- The special cases $\beta = 0$ and $\beta = \frac{1}{2}$ have interesting properties but have not been previously studied.

Special cases of weighted scores

	Power $S_{\beta}^{\mathbf{P}}(\mathbf{p}, \mathbf{e}_i \ \mathbf{q})$	Pseudospherical $S_{\beta}^{\mathbf{S}}(\mathbf{p}, \mathbf{e}_i \ \mathbf{q})$
$\beta = -1$	$-\frac{1}{2}(1 + (q_i/p_i)^2) + E_{\mathbf{q}}[\mathbf{q}/\mathbf{p}]$	$\frac{1}{2}(1 - ((q_i/p_i)/E_{\mathbf{q}}[\mathbf{q}/\mathbf{p}]))^2$
$\beta = 0$	$1 - (q_i/p_i) + E_{\mathbf{q}}[\ln(\mathbf{q}/\mathbf{p})]$	$1 - (q_i/p_i) \exp(-E_{\mathbf{q}}[\ln(\mathbf{q}/\mathbf{p})])$
$\beta = \frac{1}{2}$	$2 \left(2 - \sqrt{q_i/p_i} - E_{\mathbf{p}}[\sqrt{\mathbf{q}/\mathbf{p}}] \right)$	$2 \left(1 - \sqrt{q_i/p_i} E_{\mathbf{p}}[\sqrt{\mathbf{q}/\mathbf{p}}] \right)$
$\beta = 1$	$\ln(p_i/q_i)$	$\ln(p_i/q_i)$
$\beta = 2$	$((p_i/q_i) - 1) - \frac{1}{2} (E_{\mathbf{p}}[\mathbf{p}/\mathbf{q}] - 1)$	$((p_i/q_i)/\sqrt{E_{\mathbf{p}}[\mathbf{p}/\mathbf{q}]} - 1)$

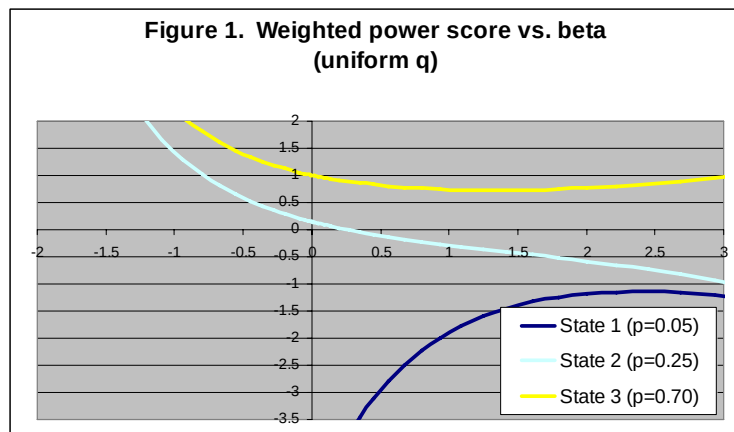
Weighted *expected* score functions

- Weighted *power* expected score:

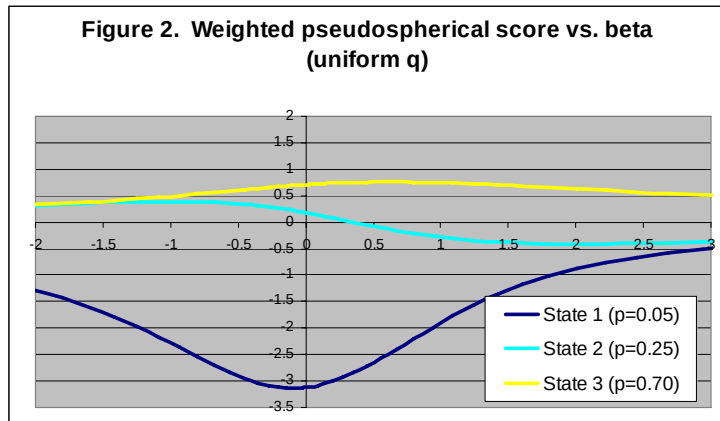
$$S_{\beta}^{\mathbf{P}}(\mathbf{p}||\mathbf{q}) = \frac{E_{\mathbf{P}}[(\mathbf{p}/\mathbf{q})^{\beta-1}] - 1}{\beta(\beta - 1)}$$

- Weighted *pseudospherical* expected score:

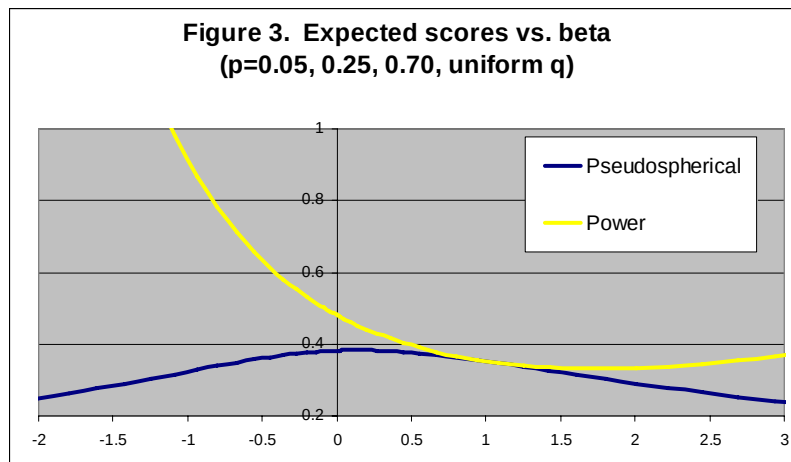
$$S_{\beta}^{\mathbf{S}}(\mathbf{p}||\mathbf{q}) = \frac{(E_{\mathbf{P}}[(\mathbf{p}/\mathbf{q})^{\beta-1}])^{1/\beta} - 1}{\beta - 1}$$



- Behavior of the weighted power score for $n = 3$.
- For fixed $\mathbf{p}$ and $\mathbf{q}$, the scores diverge as $\beta \rightarrow \pm \infty$.
- For $\beta \ll 0$ [$\beta \gg 2$] only the lowest [highest] probability event is distinguished from the others.



- By comparison, the weighted pseudospherical scores approach fixed limits as $\beta \rightarrow \pm \infty$.
- Again, for $\beta \ll 0$ [$\beta \gg 2$] only the lowest [highest] probability event is distinguished from the others.



The corresponding expected scores vs. β are equal at $\beta = 1$, where both rules converge to the weighted logarithmic scoring rule, but elsewhere the weighted power expected score is strictly larger.

Part 2. Entropy

- (An) *entropy* is a function of a probability distribution that measures its degree of “uninformativeness” or the amount of “disorder” in a physical system whose internal states have that distribution.
- (An) entropy is typically maximized by the uniform distribution, and minimized by a degenerate distribution that assigns probability 1 to a single state.

Physical entropy

- In statistical physics, the entropy of a system with n possible internal states having probability distribution $\mathbf{p}$ is defined (up to a multiplicative constant) by

$$H(\mathbf{p}) \equiv \sum_i p_i \ln(1/p_i)$$

- This statistical definition agrees with the definition of entropy in the classical thermodynamics: it is a quantity that increases whenever energy is redistributed within or between physical systems.

Information entropy

- In communication theory, under optimal encoding, the number of bits required to transmit the occurrence of an event with probability p_i from a stationary random process is (proportional to) $\ln(1/p_i)$. (Shannon 1948)
- Hence the physicist's entropy measures the average number of bits per event ("bandwidth") that is needed to communicate the state which has occurred.
- Greater entropy (larger required bandwidth) means less a priori information about the state.

The KL divergence

- The negative entropy $-H(\mathbf{p})$ therefore measures the information already possessed concerning a state whose distribution is $\mathbf{p}$.
- The *cross-entropy*, or *Kullback-Leibler divergence*, between *two* distributions is defined by

$$D^{KL}(\mathbf{p}, \mathbf{q}) \equiv \sum_i p_i (\ln(1/q_i) - \ln(1/p_i)) = \mathbf{E}_{\mathbf{p}}[\ln(\mathbf{p}/\mathbf{q})]$$

- It measures the information gained (reduction in average number of bits per event) when an initial distribution $\mathbf{q}$ distribution is replaced by an updated distribution $\mathbf{p}$ and coding is re-optimized.

Properties of the KL divergence

- *Additivity* with respect to independent partitions of the state space:

$$D^{KL}(\mathbf{p}_A \times \mathbf{p}_B; \mathbf{q}_A \times \mathbf{q}_B) = D^{KL}(\mathbf{p}_A; \mathbf{q}_A) + D^{KL}(\mathbf{p}_B; \mathbf{q}_B)$$

- Thus, if A and B are independent events whose initial distributions $\mathbf{q}_A$ and $\mathbf{q}_B$ are respectively updated to $\mathbf{p}_A$ and $\mathbf{p}_B$, the *total* expected information gain in their product space is the sum of the separate expected information gains, as measured by their KL divergences.

Properties of the KL divergence

- *Recursivity* with respect to the splitting of events:

$$\begin{aligned} & D^{KL}(p_1, \dots, p_n; q_1, \dots, q_n) \\ = & D^{KL}(p_1 + p_2, p_3, \dots, p_n; q_1 + q_2, q_3, \dots, q_n) \\ & + (p_1 + p_2) D^{KL}\left(\frac{p_1}{p_1 + p_2}, \frac{p_2}{p_1 + p_2}; \frac{q_1}{q_1 + q_2}, \frac{q_2}{q_1 + q_2}\right) \end{aligned}$$

- Thus, the total expected information gain does not depend on whether the true state is resolved all at once or via a sequential splitting of events.

Other divergence/distance measures

- The *Chi-square divergence* (Pearson 1900) is used by frequentist statisticians to measure goodness of fit:

$$\chi^2(\mathbf{p}, \mathbf{q}) = E_{\mathbf{p}}[\mathbf{p}/\mathbf{q}] - 1$$

- The *Hellinger distance* is a symmetric measure of distance between two distributions that is also popular in statistics and computer science:

$$D^{HEL}(\mathbf{p}, \mathbf{q}) \equiv \|\sqrt{\mathbf{p}} - \sqrt{\mathbf{q}}\|_2 = \left(\sum_{j=1}^n (\sqrt{p_j} - \sqrt{q_j})^2 \right)^{1/2}$$

Onward to generalized divergence...

- The properties of additivity and recursivity can be considered as *axioms* for a measure of expected information gain which imply the KL divergence.
- However, weaker axioms of “pseudoadditivity” and “pseudorecursivity” lead to parametric families of generalized divergence.
- These generalized divergences *interpolate* and *extrapolate* beyond the KL divergence, the Chi-square divergence, and the Hellinger distance.

Power divergence

- The *directed divergence of order β* , a.k.a. the *power divergence*, was proposed by Havrda & Chavráť (1967) and elaborated by Rathie & Kannappan (1972), Cressie & Read (1980), Haussler and Oppen (1997):

$$D_{\beta}^{\mathbf{P}}(\mathbf{p}||\mathbf{q}) \equiv \frac{E_{\mathbf{P}}[(\mathbf{p}/\mathbf{q})^{\beta-1}] - 1}{\beta(\beta - 1)}$$

- It is pseudoadditive and pseudorecursive for all β , and it coincides with the KL divergence at $\beta = 1$.
- It is the weighted power expected score, hence:

The power divergence is the implicit information measure behind the weighted power scoring rule.

Pseudospherical divergence

- An alternative generalized entropy was introduced by Arimoto (1971) and further studied by Sharma & Mittal (1975), Boeke & Van der Lubbe (1980) and Lavenda & Dunning-Davies (2003), for $\beta > 1$:

$$\frac{\beta}{\beta - 1} \left(E_{\mathbf{P}}[\mathbf{p}^{\beta-1}]^{1/\beta} - 1 \right)$$

- The corresponding divergence, which we call the *pseudospherical divergence*, is obtained by introducing a baseline distribution $\mathbf{q}$ and dividing out the unnecessary β in the numerator:

$$D_{\beta}^{\mathbf{S}}(\mathbf{p}||\mathbf{q}) \equiv \frac{(E_{\mathbf{P}}[(\mathbf{p}/\mathbf{q})^{\beta-1}])^{1/\beta} - 1}{\beta - 1}$$

Properties of the pseudospherical divergence

- It is defined for all real β (not merely $\beta > 1$).
- It is pseudoadditive but generally not pseudorecursive.
- It is identical to the weighted pseudospherical expected score, hence:

The pseudospherical divergence is the implicit information measure behind the weighted pseudospherical scoring rule.

Interesting special cases

- The power and pseudospherical divergences both coincide with the KL divergence at $\beta = 1$.
- At $\beta = 0$, $\beta = 1/2$, and $\beta = 2$ they are linearly (or at least monotonically) related to the reverse KL divergence, the squared Hellinger distance, and the Chi-square divergence, respectively:

	$S_{\beta}^P(\mathbf{p} \mathbf{q}) = D_{\beta}^P(\mathbf{p} \mathbf{q})$	$S_{\beta}^S(\mathbf{p} \mathbf{q}) = D_{\beta}^S(\mathbf{p} \mathbf{q})$
$\beta = -1$	$\frac{1}{2}\chi^2(\mathbf{q} \mathbf{p})$	$\frac{1}{2} (1 - (\chi^2(\mathbf{q} \mathbf{p}) + 1)^{-1})$
$\beta = 0$	$D_{KL}(\mathbf{q} \mathbf{p})$	$1 - \exp(-D_{KL}(\mathbf{q} \mathbf{p}))$
$\beta = \frac{1}{2}$	$2D_H(\mathbf{p} \mathbf{q})^2 = 2D_H(\mathbf{q} \mathbf{p})^2$	$2 \left(1 - \left(1 - \frac{1}{2}D_H(\mathbf{p} \mathbf{q})^2 \right)^2 \right)$
$\beta = 1$	$D_{KL}(\mathbf{p} \mathbf{q})$	$D_{KL}(\mathbf{p} \mathbf{q})$
$\beta = 2$	$\frac{1}{2}\chi^2(\mathbf{p} \mathbf{q})$	$\sqrt{\chi^2(\mathbf{p} \mathbf{q}) + 1} - 1$

Where we've gotten so far...

- There are two parametric families of *weighted*, strictly proper scoring rules which correspond exactly to two well-known parametric families of *generalized divergence*, each of which has a full “spectrum” of possibilities ($-\infty < \beta < +\infty$).
- But what is the decision-theoretic significance of these quantities?
- What are some guidelines for choosing among the the two families and their parameters?

Part 3. Decisions under uncertainty with linear risk tolerance

- Suppose a decision maker with subjective probability distribution $\mathbf{p}$ and utility function u bets or trades optimally against a risk-neutral opponent or contingent claim market with distribution $\mathbf{q}$.
- For any risk-averse utility function, the investor's gain in expected utility yields an economic measure of the divergence between $\mathbf{p}$ and $\mathbf{q}$.
- In particular, suppose the investor's utility function belongs to the *linear risk tolerance* (HARA) family, i.e., the family of generalized exponential, logarithmic, and power utility functions.

Two canonical decision problems:

- **Problem “S”:** A risk averse decision maker with probability distribution $\mathbf{p}$ and utility function $u(x)$ for time-1 consumption bets optimally at time 0 against a risk-neutral opponent with distribution $\mathbf{q}$ to obtain the expected utility:

$$U^S(\mathbf{p}||\mathbf{q}) \equiv \max_{\mathbf{x} \in \mathbf{R}^n} \underbrace{E_{\mathbf{p}}[u(\mathbf{x})]}_{\text{Decision maker's expected utility}} \quad \text{s.t.} \quad \underbrace{E_{\mathbf{q}}[\mathbf{x}] \leq 0}_{\text{Feasibility constraint: } -\mathbf{x} \text{ must have non-negative expected value for opponent}}$$

Decision maker's payoff vector

Two canonical problems, continued:

- **Problem “P”:** A risk averse decision maker with distribution $\mathbf{p}$ and *quasilinear* utility function $a + u(b)$, where a is time-0 consumption and b is time-1 consumption, bets optimally at time 0 against a risk-neutral opponent with distribution $\mathbf{q}$, to obtain the expected utility:

$$U^P(\mathbf{p}||\mathbf{q}) \equiv \max_{\mathbf{x} \in \mathbf{R}^n} \underbrace{E_{\mathbf{p}}[u(\mathbf{x})]}_{\text{Expected utility gained at time 1}} - \underbrace{E_{\mathbf{q}}[\mathbf{x}]}_{\text{Utility lost at time 0 (cost of } \mathbf{x})}$$

Decision maker's time-1 payoff vector

Risk aversion and risk tolerance

- Let $u(x)$ denote the vNM utility of wealth x .
- The monetary quantity $\tau(x) \equiv -u'(x)/u''(x)$ is the investor's local *risk tolerance* at x (the reciprocal of the Pratt-Arrow measure of local risk aversion).
- The usual decision-analytic rule of thumb is that your local risk tolerance is roughly the monetary amount τ such that you are just indifferent to accepting an objective 50-50 gamble with payoffs $+\tau$ and $-\frac{1}{2}\tau$.
- For example, your risk tolerance is \$10,000 if you are just indifferent to accepting a coin-flip gamble between winning \$10,000 and losing \$5,000.

Linear risk tolerance (LRT) utility

- The most commonly used utility functions in decision analysis and financial economics have the property of *linear risk tolerance*, i.e., $\tau(x) = \alpha + \beta x$, where β is the *risk tolerance coefficient*.
- Henceforth let x denote gain or loss relative to a riskless status quo, and w.l.o.g. choose the unit of money and the origin and scale of utility so that $u(0) = 0$ and $u'(0) = 1$ and $\tau(x) = 1 + \beta x$, yielding the normalized functional form:
- $$u_{\beta}(x) \equiv \frac{1}{\beta - 1}((1 + \beta x)^{(\beta-1)/\beta} - 1) \text{ if } \beta x > -1,$$

$$\equiv -\infty \text{ otherwise,}$$

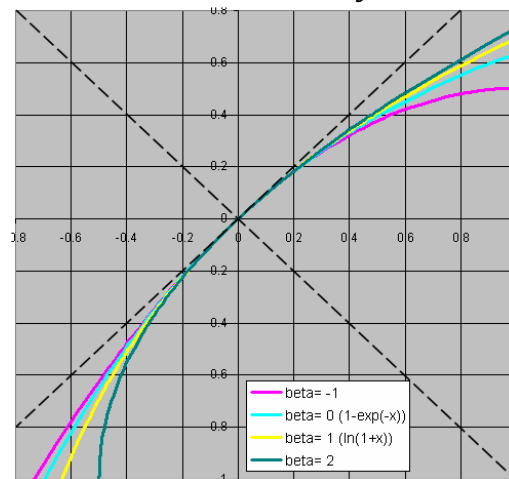
Special cases of normalized LRT utility

$\beta = -1$	quadratic utility	$u_{-1}(x) = -\frac{1}{2}((1-x)^2 - 1)$
$\beta = 0$	exponential utility	$u_0(x) = 1 - \exp(-x)$
$\beta = \frac{1}{2}$	reciprocal utility	$u_{1/2}(x) = 2 \left(1 - \frac{1}{1+x/2}\right)$
$\beta = 1$	logarithmic utility	$u_1(x) = \ln(1+x)$
$\beta = 2$	square-root utility	$u_2(x) = \sqrt{1+2x} - 1$

Note the symmetry around $\beta = \frac{1}{2}$. Also note that for $\beta < 0$ (e.g., quadratic utility) risk tolerance counterintuitively *decreases* with wealth (“increasing absolute risk aversion”), while for $\beta > 1$ (e.g., square root utility), risk tolerance counterintuitively *increases faster* than wealth (“decreasing relative risk aversion”).

Qualitative properties of LRT utility

The graphs of $u_\beta(x)$ and $u_{1-\beta}(x)$, whose powers are reciprocal to each other, are reflections of each other around the line $y = -x$.



Reciprocal $\leftrightarrow$ Reciprocal

Exponential $\leftrightarrow$ Log

Quadratic $\leftrightarrow$ Square root

Power $\delta \leftrightarrow$ Power $1/\delta$

...where $\delta = (\beta - 1)/\beta$

First main result

Let $\mathbf{x}_\beta^{\mathbf{S}}(\mathbf{p}||\mathbf{q})$ and $\mathbf{x}_\beta^{\mathbf{P}}(\mathbf{p}||\mathbf{q})$ denote the solutions of Problems **S** and **P**, with i^{th} elements $x_{\beta,i}^{\mathbf{S}}(\mathbf{p}||\mathbf{q})$ and $x_{\beta,i}^{\mathbf{P}}(\mathbf{p}||\mathbf{q})$, respectively.

Then the utility gains to the decision maker under problems **S** and **P** are the pseudospherical and power scores for the same $\mathbf{p}$, $\mathbf{q}$, and β , and the weighted power expected score is greater than or equal to the corresponding weighted pseudospherical expected score:

THEOREM 1:

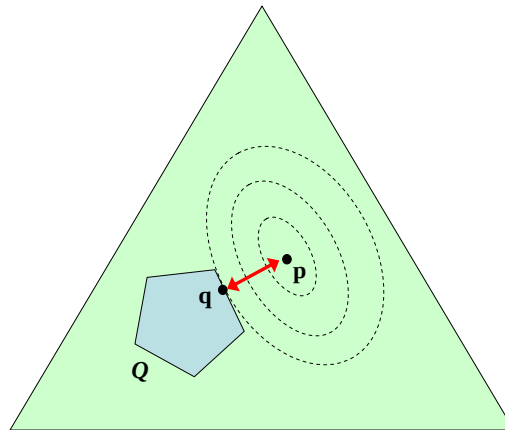
- (a) $S_\beta^{\mathbf{S}}(\mathbf{p}, \mathbf{e}_i||\mathbf{q}) = u_\beta(x_{\beta,i}^{\mathbf{S}}(\mathbf{p}||\mathbf{q}))$, and $S_\beta^{\mathbf{S}}(\mathbf{p}||\mathbf{q}) = U_\beta^{\mathbf{S}}(\mathbf{p}||\mathbf{q})$
- (b) $S_\beta^{\mathbf{P}}(\mathbf{p}, \mathbf{e}_i||\mathbf{q}) = u_\beta(x_{\beta,i}^{\mathbf{P}}(\mathbf{p}||\mathbf{q})) - E_{\mathbf{q}}[x_\beta^{\mathbf{P}}(\mathbf{p}||\mathbf{q})]$, and $S_\beta^{\mathbf{P}}(\mathbf{p}||\mathbf{q}) = U_\beta^{\mathbf{P}}(\mathbf{p}||\mathbf{q})$
- (c) $S_\beta^{\mathbf{P}}(\mathbf{p}||\mathbf{q}) \geq S_\beta^{\mathbf{S}}(\mathbf{p}||\mathbf{q})$ for all $\mathbf{p}$, $\mathbf{q}$, and β .

Extension to imprecise probabilities/incomplete markets

- Suppose the decision maker faces a risk neutral opponent with imprecise probabilities (or an *incomplete* market) whose beliefs (prices) determine only a convex set Q of probability distributions.
- Then the utility-maximization problems **S** and **P** generalize into convex programs whose duals are the minimization of the corresponding divergences (expected scores).

Generalization of problem S

- A payoff vector $\mathbf{x}$ is feasible for the decision maker if the opponent's (market's) payoff $-\mathbf{x}$ has non-negative expectation for every $\mathbf{q}$ in Q .
- **Primal problem:** Find $\mathbf{x}$ in $\mathcal{R}^n$ to maximize $E_{\mathbf{p}}[u_{\beta}(\mathbf{x})]$ subject to $E_{\mathbf{q}}[\mathbf{x}] \leq \mathbf{0}$ for all $\mathbf{q}$ in Q .
- **Dual problem:** Find $\mathbf{q}$ in Q that minimizes $S_{\beta}^{\mathbf{S}}(\mathbf{p} \parallel \mathbf{q})$, the pseudospherical divergence from $\mathbf{p}$.



Generalization of problem S:

$\mathbf{p}$ = precise probability of a risk averse decision maker with utility function u

Q = set of imprecise probabilities of risk neutral opponent/market

Finding the payoff vector $\mathbf{x}$ to maximize $E_{\mathbf{p}}[u(\mathbf{x})]$ s.t. $E_{\mathbf{q}}[\mathbf{x}] \leq 0$ is equivalent (dual) to finding $\mathbf{q}$ in Q to minimize the divergence $S(\mathbf{p} \parallel \mathbf{q})$

Generalization of problem P

- A time-1 payoff vector $\mathbf{x}$ can be purchased for a price w at time 0 if the opponent's (market's) payoff $-\mathbf{x}$ has an expected value of at least w for every $\mathbf{q}$ in Q
- **Primal problem:** Find $\mathbf{x}$ in $\mathfrak{R}^n$ to maximize $E_{\mathbf{p}}[u_{\beta}(\mathbf{x})] - w$ subject to $E_{\mathbf{q}}[\mathbf{x}] \leq w$ for all $\mathbf{q}$ in Q .
- **Dual problem:** Find $\mathbf{q}$ in Q that minimizes $S_{\beta}^{\mathbf{P}}(\mathbf{p} \parallel \mathbf{q})$, the power divergence from $\mathbf{p}$.

Conclusions (i)

- The power & pseudospherical scoring rules can (and should) be generalized by incorporating a not-necessarily-uniform baseline distribution.
- The resulting weighted *expected* scores are equal to well-known generalized divergences, with KL divergence as the special case $\beta = 1$.
- These scoring rules and divergences also arise as the solutions to utility maximization problems with LRT utility in 1 period or quasilinear LRT utility in 2 periods, where the baseline distribution describes the beliefs of a risk neutral betting opponent (or market)

Conclusions (ii)

- When the baseline distribution is *imprecise* (market incompleteness), the problem of maximizing expected utility is the *dual* of the problem of minimizing the corresponding divergence.
- These results shed more light on the connection between utility theory and information theory, particularly with respect to commonly-used parametric forms of utility and divergence.
- For the weighted power and pseudospherical scoring rules, values of β between 0 and 1 appear to be the most interesting (because they correspond to intuitively reasonable risk tolerance coefficients in Problems **S** & **P**), and the cases $\beta = 0$ and $\beta = \frac{1}{2}$ have been so far under-explored.